Chapter 4
Model Adequacy Checking

The fitting of the linear regression model, estimation of parameters, testing of hypotheses, and properties of the estimator are based on the following major assumptions:

1. The relationship between the study variable and explanatory variables is linear, at least approximately.
2. The error term has zero mean.
3. The error term has a constant variance.
4. The errors are uncorrelated.
5. The errors are normally distributed.

The validity of these assumptions is needed for the results to be meaningful. If these assumptions are violated, the result can be incorrect and may have serious consequences. If these departures are small, the final result may not be changed significantly. But if the deviations are large, the model obtained may become unstable, in the sense that a different sample could yield an entirely different model with opposite conclusions. So, such underlying assumptions must be verified before attempting regression modelling. Such information is not available from summary statistics, such as the *t*-statistic, *F*-statistic, or coefficient of determination.

One crucial point to keep in mind is that these assumptions apply to the population, but we work only with a sample. The main issue is deciding about the population based on a sample.

Several diagnostic methods for checking violations of regression assumptions rely on examining model residuals using various types of graphics.

Checking the linear relationship between the study and explanatory variables

1. Case of one explanatory variable

If there is only one explanatory variable in the model, then it is easy to check the existence of a linear relationship between y and X by a scatter diagram of the available data.

If the scatter diagram shows a linear trend, it indicates that the relationship between y and X is linear. If the pattern is not linear, it suggests that the relationship between y and X is nonlinear. For example, the following figure 1 indicates a linear trend

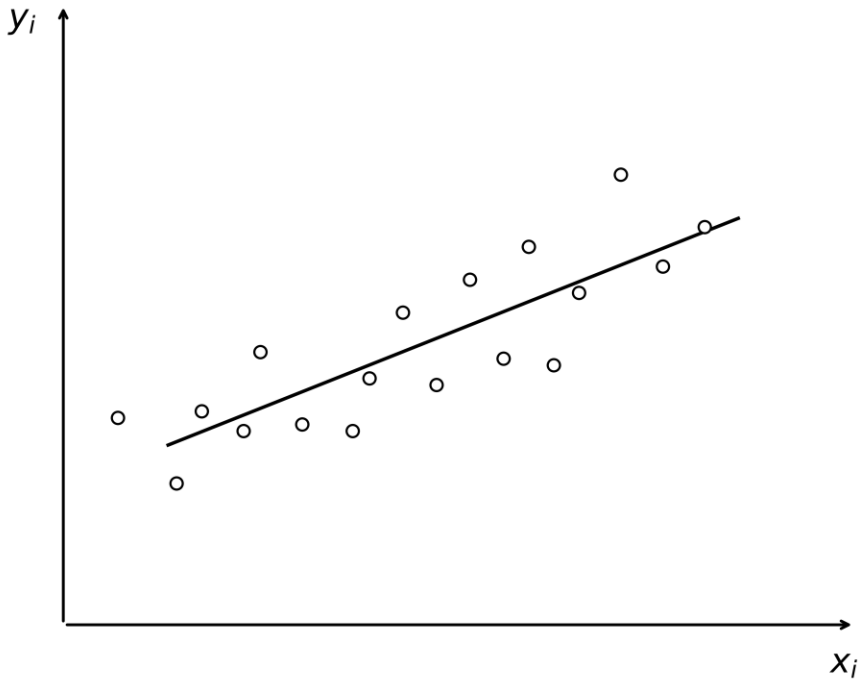


Figure 1: Linear trend

whereas the following graph in Figure 2 suggests a nonlinear trend:

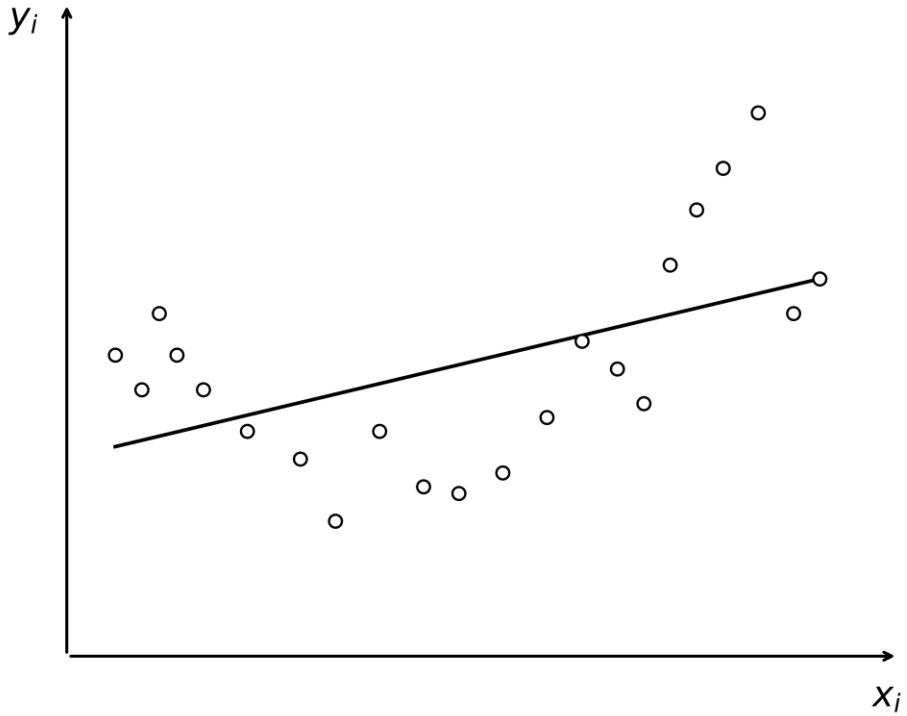


Figure 2 Nonlinear trend

2. Case of more than one explanatory variable

To check the assumption of linearity between the study variable and the explanatory variables, a **scatter plot matrix** of the data can be used. A scatterplot matrix is a two-dimensional array of scatter plots, with each plot except the diagonal containing a scatter diagram. Thus, each scenario sheds some light on the relationship between a pair of variables. It provides more information than the correlation coefficient for each pair of variables, because it indicates whether the relationship is linear or nonlinear and provides a sense of how the individual data points are arranged across the region. It is a scatter diagram of $(y \text{ versus } X_1)$, $(y \text{ versus } X_2)$, …, $(y \text{ versus } X_k)$.

Another option to present the scatterplot is

- display the scatterplots in the upper triangular part of the plot matrix.
- Mention the corresponding correlation coefficients in the lower triangular part of the matrix.

Suppose there are only two explanatory variables and the model is $y = X_1\beta_1 + X_2\beta_2 + \varepsilon$, then the scatterplot matrix looks like as follows in Figure 3.

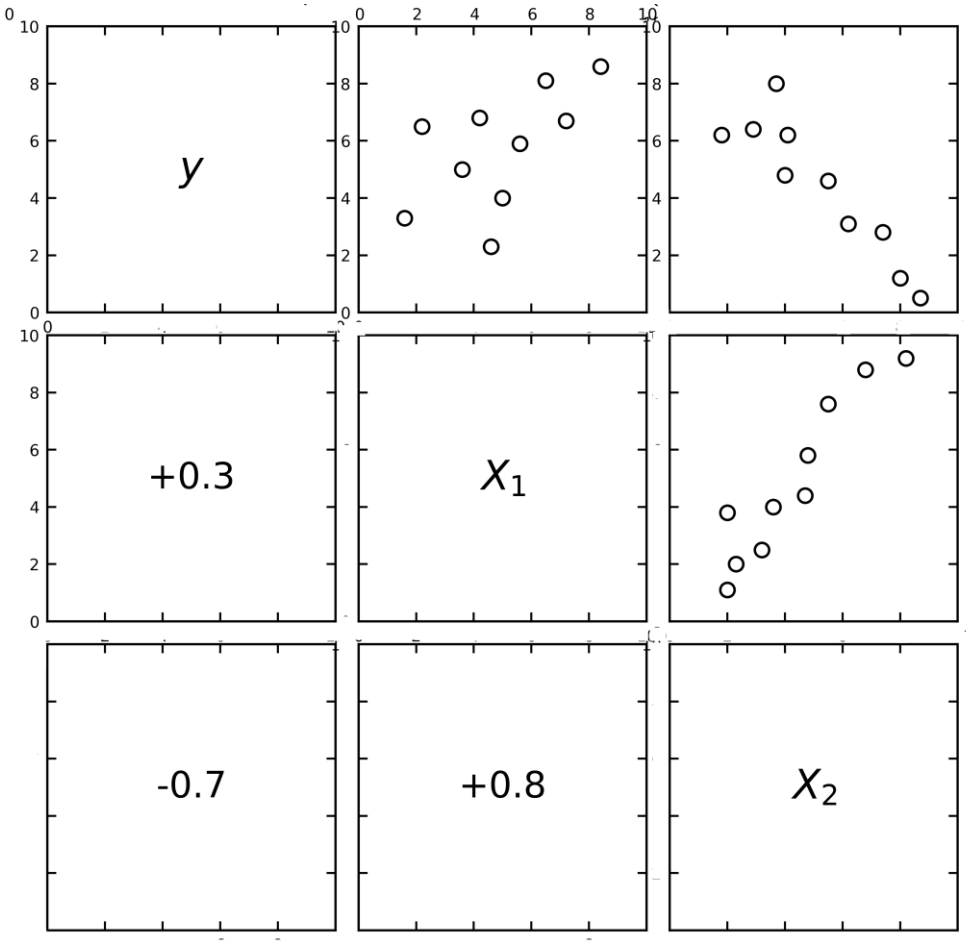


Figure 3: Scatterplot matrix

Such an arrangement helps in examining the plot and the corresponding correlation coefficient together. The pairwise correlation coefficient should always be interpreted in conjunction with the corresponding scatter plots because

- the correlation coefficient measures only the linear relationship and
- the correlation coefficient is non-robust, i.e., one or two observations can substantially influence its value in the data.

The presence of linear patterns is reassuring, but the absence of such patterns does not imply that the linear model is incorrect. Most statistical software provides the option to create a scatterplot matrix. The plots indicate that a multiple linear regression model may provide a reasonable fit to the data. It is to be kept in mind that we get only the information on pairs of variables through the scatterplot of $(y \text{ versus } X_1)$, $(y \text{ versus } X_2)$, …, $(y \text{ versus } X_k)$ whereas the assumption of linearity is between y and jointly with $(X_1, X_2, .., X_k)$.

If some explanatory variables are interrelated, these scatter diagrams can be misleading. Other methods for sorting out the relationships among several explanatory variables and a study variable are used.

Residual analysis

The **residual** is defined as the difference between the observed and fitted value of the study variable. The ith residual is defined as

$$e_i = y_i \sim \hat{y}_i = y_i - \hat{y}_i,\ i = 1, 2, ..., n$$

where y_i is an observation and $\hat{y}_i$ is the corresponding fitted value.

Residuals can be viewed as the deviations between the data and the fit. So it is also a measure of the variability in the response variable that is not explained by the regression model.

Residuals can be thought of as the observed values of the model errors. So it can be expected that any departure from the assumptions about random errors should show up in the residuals. The analysis of residuals helps identify model inadequacies.

Assuming that the OLSE estimates the regression coefficients in the model $y = X\beta + \varepsilon$, we find that:

- Residuals have a zero mean as

$$\begin{aligned} E(e_i) &= E(y_i - \hat{y}_i) \\ &= E(X_i\beta + \varepsilon_i - X_i b) \\ &= X_i\beta + 0 - X_i\beta \\ &= 0 \end{aligned}$$

- An approximate average variance of residuals is estimated by

$$\frac{\sum_{i=1}^{n}(e_i - \bar{e})^2}{n-k} = \frac{\sum_{i=1}^{n} e_i^2}{n-k} = \frac{SS_{res}}{n-k} = MS_{res}.$$

- Residuals are not independent, as the n residuals have only $n-k$ degrees of freedom. The nonindependence of the residuals has little effect on their use for model adequacy checking as long as n is not small relative to k.

Methods for scaling residuals

Sometimes it is easier to work with scaled residuals. We discuss four ways for scaling the residuals.

1. Standardized residuals

The residuals are standardized based on the concept of residual minus its mean and divided by its standard deviation. Since $E(e_i) = 0$ and MS_{res} estimates the approximate average variance, so logically, the scaling of the residual is

$$d_i = \frac{e_i}{\sqrt{MS_{res}}},\ i = 1, 2, ..., n$$

is called as standardized residual for which

$$\begin{aligned} E(d_i) &= 0 \\ Var(d_i) &\approx 1. \end{aligned}$$

So a large value of $d_i (> 3,$ say) potentially indicates an outlier.

2. Studentized residuals

The standardized residuals use the approximate variance of e_i as MS_{res}. The studentized residuals use the exact variance of e_i.

We first find the variance of e_i.

In the model $y = X\beta + \varepsilon$, the OLSE of β is $b = (X'X)^{-1}X'y$ and the residual vector is

$$\begin{aligned} e &= y - \hat{y} \\ &= y - Xb \\ &= y - Hy \\ &= (I - H)y \quad \text{where } H = X(X'X)^{-1}X' \\ &= (I - H)(X\beta + \varepsilon) \\ &= X\beta - HX\beta + (I - H)\varepsilon \\ &= X\beta - X\beta + (I - H)\varepsilon \\ &= (I - H)\varepsilon \\ &= \bar{H}\varepsilon \end{aligned}$$

Thus $e = \bar{H}y = \bar{H}\varepsilon$, so the residuals are the same linear transformation of y and ε.

The covariance matrix of residuals is

$$\begin{aligned} V(e) &= V(\bar{H}\varepsilon) \\ &= \bar{H}V(\varepsilon)\bar{H} \\ &= \sigma^2\bar{H} \\ &= \sigma^2(I - H) \end{aligned}$$

and $V(\varepsilon) = \sigma^2 I.$

The matrix $(I - H)$ is symmetric and idempotent but generally not diagonal. So the residuals have different variances and are correlated.

If h_{ii} is the ith diagonal element of the hat matrix H and h_{ij} is the $(i, j)^{th}$ element of H, then

$$\begin{aligned} Var(e_i) &= \sigma^2(1 - h_{ii}) \\ Cov(e_i, e_j) &= -\sigma^2 h_{ij}. \end{aligned}$$

Since $0 \le h_{ii} \le 1$, so if h_{ii} is small, then $Var(e_i)$ is larger, which indicates a poorer fit. If MS_{res} is used to estimate the $Var(e_i)$ then

$$\begin{aligned} \widehat{Var}(e_i) &= \hat{\sigma}^2(1 - h_{ii}) \\ &= MS_{res}(1 - h_{ii}) \end{aligned}$$

$\Rightarrow MS_{res}$ overestimates the $Var(e_i)$.

Now we discuss that h_{ii} is a measure of the location of the ith point in x-space.

Regression variable hull (RVH)

A regression variable hull is the smallest convex set containing all the original data $x_i = (x_{i1}, x_{i2}, ..., x_{ik}),\ i = 1, 2, ..., n.$

A pictorial presentation of a RVH is as follows in Figure 4 for better understanding:

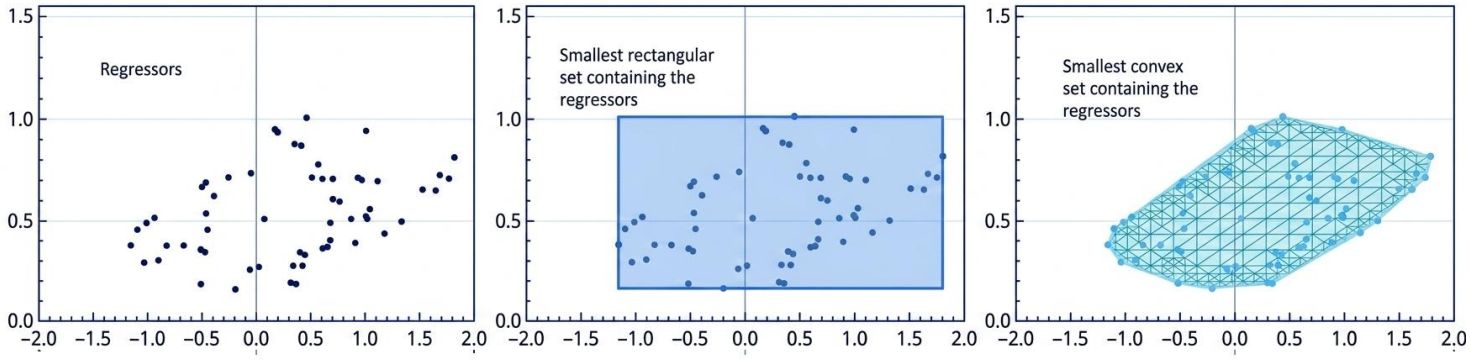


Figure 4: Scatter diagram and regression variable hull

The $h_{ii} = x_i (X'X)^{-1} x_i'$ is a measure of the location of the ith point in x-space and depends on the Euclidean distance of x_i from the centroid and on the density of the points in RVH.

In general, if a point has the largest value of h_{ii}, say $h_{\max}$, then it will lie on the boundary of the RVH in a region of the x-space. In such a region, where the density of the observations is relatively low. The set of points x (not necessarily the data points used to fit the model) that satisfy

$$x'(X'X)^{-1} x \le h_{\max}$$

is an ellipsoid enclosing all points inside the RVH. So the location of a point, say, $x_0 = (x_{01}, x_{02}, ..., x_{0k})$, relative to RVH is given by

$$h_{00} = x_0'(X'X)^{-1} x_0,$$

and the points for which $h_{00} > h_{\max}$ are outside the ellipsoid containing RVH. If $h_{00} < h_{\max}$ then the point is inside the RVH. Generally, the smaller the value of h_{00} indicates that the point x_0 lies closer to the centroid of the x-space.

Since h_{ii} is a measure of the location of the ith point in x-space, the variance of e_i depends on where the point x_i lies. If h_{ii} is small, then $Var(e_i)$ is larger, which indicates a poorer fit. So the points near the centre of the x-space have poorer least-squares fit than the residuals at more remote locations. Violations of model

assumptions are more likely at distant points, and these violations may be hard to detect from the inspection of ordinary residuals e_i (or the standardized residuals d_i) because their residuals will usually be smaller.

So a logical procedure is to examine the studentized residuals of the form

$$r_i = \frac{e_i}{\sqrt{MS_{res}(1-h_{ii})}}$$

in place of e_i (or d_i). For r_i,

$$E(r_i) = 0$$
$$Var(r_i) = 1$$

regardless of the location of x_i when the form of the model is correct.

In many situations, the variance of residuals stabilizes (particularly in large data sets), and there may be little difference between d_i and r_i. In such cases, d_i and r_i often convey equivalent information.

However, since any point with a

- large residual and
- large h_{ii}

is potentially highly influential on the least-squares fit, so examination of r_i is generally recommended.

If there is only one explanatory variable, then

$$r_i = \frac{e_i}{\sqrt{MS_{res}\left[1-\left(\frac{1}{n}+\frac{(x_i-\bar{x})^2}{s_{xx}}\right)\right]}},\ i = 1,2,...,n\ .$$

- When x_i is close to the midpoint of x-data, i.e., $x_i - \bar{x}$ is small then estimated standard deviation of e_i is large.
- Conversely, when x_i is near the extreme ends of the range of x-data, then $x_i - \bar{x}$ is large and estimated standard deviation of e_i is small.
- When n is really large, the effect of $(x_i - \bar{x})^2$ is relatively small. So in big data sets, r_i may not differ dramatically from d_i.

3. PRESS residuals

The PRESS residuals are defined as $(y_i - \hat{y}_{(i)})$ where $\hat{y}_{(i)}$ is the fitted value of the ith response based on all the observations except the ith one.

Reason: If y_i is really unusual, then the regression model based on all observations may be overly influenced by it. This could produce a $\hat{y}_i$ that is very similar to y_i and consequently e_i will be small. So it will be challenging to detect any outliers.

If y_i is deleted, then $\hat{y}_{(i)}$ cannot be influenced by that observation, so the resulting residual should be likely to indicate the presence of the outlier.

Procedure

- Delete the ith observation,
- Fit the regression model to the remaining $(n-1)$ observations,
- Calculate the predicted value of y_i corresponding to the deleted observation.
- The corresponding prediction error $e_{(i)} = y_i - \hat{y}_{(i)}$
- Calculate $e_{(i)}$ for each $i = 1, 2, ..., n$.

These prediction errors are called **PRESS residuals** because they are used in computing the prediction error sum of squares. They are also called as **deleted residuals**.

Now we establish a relationship between e_i and $e_{(i)}$.

Relation between e_i and $e_{(i)}$

Let $b_{(i)}$ be the vector of regression coefficients estimated by withholding the ith observations. Then

$$b_{(i)} = \left(X_{(i)}' X_{(i)}\right)^{-1} X_{(i)}' y_{(i)}$$

where $X_{(i)}$ is the X-matrix without the vector of ith observation and $y_{(i)}$ is the y-vector without the ith observation. Then

$$\begin{aligned} e_{(i)} &= y_i - \hat{y}_{(i)} \\ &= y_i - x_i \hat{b}_{(i)} \\ &= y_i - x_i (X_{(i)}' X_{(i)})^{-1} X_{(i)}' y_{(i)} \end{aligned}$$

We use the following result in further analysis.

Result: If $X'X$ is a $k \times k$ matrix and x_i be its ith row vector then $(X'X - x_i'x_i)$ denotes the $X'X -$ matrix with the ith row withheld. Then

$$\left[X'X - x_i'x_i\right]^{-1} = \left[X_{(i)}'X_{(i)}\right]^{-1} = (X'X)^{-1} + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}}{1 - x_i(X'X)^{-1}x_i'}$$
$$= (X'X)^{-1} + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}}{1 - h_{ii}}.$$

where $h_{ii} = x_i(X'X)^{-1}x_i'$.

Then using this result, we can write

$$\begin{aligned}
e_{(i)} &= y_i - x_i\left(X_{(i)}'X_{(i)}\right)^{-1}X_{(i)}'y_{(i)} \\
&= y_i - x_i\left[(X'X)^{-1} + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}}{1-h_{ii}}\right]X_{(i)}'y_{(i)} \\
&= y_i - x_i(X'X)^{-1}X_{(i)}'y_{(i)} - \frac{x_i(X'X)^{-1}x_i'x_i(X'X)^{-1}X_{(i)}'y_{(i)}}{1-h_{ii}} \\
&= y_i - x_i(X'X)^{-1}X_{(i)}'y_{(i)} - \frac{h_{ii}x_i(X'X)^{-1}X_{(i)}'y_{(i)}}{1-h_{ii}} \\
&= \frac{(1-h_{ii})y_i - (1-h_{ii})x_i(X'X)^{-1}X_{(i)}'y_{(i)} - h_{ii}x_i(X'X)^{-1}X_{(i)}'y_{(i)}}{1-h_{ii}} \\
&= \frac{(1-h_{ii})y_i - x_i(X'X)^{-1}X_{(i)}'y_{(i)}}{1-h_{ii}}.
\end{aligned}$$

Using $X'y = X_{(i)}'y_{(i)} + x_i'y_i$ (as x_i is $1 \times k$ vector), we can write

$$\begin{aligned}
e_{(i)} &= \frac{(1-h_{ii})y_i - x_i(X'X)^{-1}(X'y - x_i'y_i)}{1-h_{ii}} \\
&= \frac{(1-h_{ii})y_i - x_i(X'X)^{-1}X'y + x_i(X'X)^{-1}x_i'y_i}{1-h_{ii}} \\
&= \frac{(1-h_{ii})y_i - x_ib + h_{ii}y_i}{1-h_{ii}} \\
&= \frac{y_i - x_ib}{1-h_{ii}} \\
&= \frac{e_i}{1-h_{ii}}.
\end{aligned}$$

Looking at the relationship between e_i and $e_{(i)}$, it is clear that calculating the PRESS residuals does not require fitting in different regressions. The $e_{(i)}'s$ are just the ordinary residuals weighted according to the diagonal elements h_{ii} of H. It is possible to calculate the PRESS residuals from the residuals of a single least-squares fit to all n observations.

Residuals associated with points for which h_{ii} is large will have large PRESS residuals. Such points will generally be **high-influence** points.

The large difference between ordinary residual and PRESS residuals indicates a point where the model fits the data well, and a model without that point **predicts** poorly.

Now

$$\begin{aligned} Var(e_{(i)}) &= Var\left(\frac{e_i}{1-h_{ii}}\right) \\ &= \frac{1}{(1-h_{ii})^2} Var(e_i) \\ &= \frac{1}{(1-h_{ii})^2}(1-h_{ii})\sigma^2 \\ &= \frac{\sigma^2}{1-h_{ii}}. \end{aligned}$$

The **standardized PRESS residual** is

$$\frac{e_{(i)}}{\sqrt{Var(e_{(i)})}} = \frac{\left(\frac{e_i}{1-h_{ii}}\right)}{\sqrt{\frac{\sigma^2}{(1-h_{ii})}}} = \frac{e_i}{\sqrt{\sigma^2(1-h_{ii})}}$$

which is the same as the Studentized residuals when σ^2 is estimated by MS_{res}.

4. *R*-student

The studentized residual r_i is often considered as an outlier diagnostic, and MS_{res} is used as an estimate of σ^2 in computing r_i. This is referred to as **internal scaling** of the residuals because MS_{res} is an internally generated estimate of σ^2 obtained from fitting the model to all n observation.

Another approach is to use an estimate of σ^2 based on a data set with i^{th} observation removed, say $s^2_{(i)}$.

First, we derive an expression for $s^2_{(i)}$. Using the identity

$$\left[X'_{(i)}X_{(i)}\right]^{-1} = (X'X)^{-1} + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}}{1-h_{ii}}.$$

Post multiply both sides by $(X'y - x_i'y_i)$, we get

$$\left(X'_{(i)}X_{(i)}\right)^{-1}(X'y - x_i'y_i) = (X'X)^{-1}(X'y - x_i'y_i)$$

$$\left(X'_{(i)}X_{(i)}\right)^{-1}X'y - \left(X'_{(i)}X_{(i)}\right)^{-1}x_i'y_i = (X'X)^{-1}X'y - (X'X)^{-1}x_i'y_i + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}(X'y - x_i'y_i)}{1-h_{ii}}$$

$$\left(X'_{(i)}X_{(i)}\right)^{-1}(X'_{(i)}y_{(i)} + x_i'y_i) - \left(X'_{(i)}X_{(i)}\right)^{-1}x_i'y_i = (X'X)^{-1}X'y - (X'X)^{-1}x_i'y_i + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}(X'y - x_i'y_i)}{1-h_{ii}}$$

$$b_{(i)} = b - (X'X)^{-1}x_i'y_i + \frac{(X'X)^{-1}x_i'x_i(X'X)^{-1}(X'y - x_i'y_i)}{1-h_{ii}}$$

$$b - b_{(i)} = (X'X)^{-1}x_i'y_i - \frac{(X'X)^{-1}x_i'x_i\left[b - (X'X)^{-1}x_i'y_i\right]}{1-h_{ii}}$$

$$= \frac{(1-h_{ii})(X'X)^{-1}x_i'y_i - (X'X)^{-1}x_i'x_ib + (X'X)^{-1}x_i'h_{ii}y_i}{1-h_{ii}}$$

$$= \frac{(X'X)^{-1}x_i'\left[y_i - x_ib\right]}{1-h_{ii}}$$

$$= \frac{(X'X)^{-1}x_i'e}{1-h_{ii}}$$

$$b_{(i)} = b - \frac{(X'X)^{-1}x_i'e}{1-h_{ii}}.$$

Now consider

$$
\begin{aligned}
(n-k-1)s_{(i)}^2 &= \sum_{j\neq i=1}^{n}(y_j - x_j b_{(i)})^2 \\
&= \sum_{j=1}^{n}\left[y_j - x_j b + \frac{x_j(X'X)x_i' e_i}{1-h_{ii}} \right]^2 - \left(y_i - x_i b + \frac{h_{ii}e_i}{1-h_{ii}} \right)^2 \\
&= \sum_{j=1}^{n}\left[e_j + \frac{h_{ij}e_i}{1-h_{ii}} \right]^2 - \frac{e_i^2}{(1-h_{ii})^2} \\
&= \sum_{j=1}^{n} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j=1}^{n} e_j h_{ij} + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j=1}^{n} h_{ij}^2 - \frac{e_i}{(1-h_{ii})} \\
&= \sum_{j=1}^{n} e_j^2 + \frac{h_{ii}e_i^2}{(1-h_{ii})^2} - \frac{e_i^2}{(1-h_{ii})^2} \\
&= \sum_{j=1}^{n} e_j^2 - \frac{e_i^2}{1-h_{ii}} \quad (\text{using } Hy = H\hat{y},\ \sum_{j=1}^{n} e_j h_{ij} = 0,\ \sum_{j=1}^{n} h_{ij}^2 = h_{ij} \text{ as } H \text{ is idempotent}) \\
&= (n-k)MS_{res} - \frac{e_i^2}{1-h_{ii}}
\end{aligned}
$$

Thus

$$
s_{(i)}^2 = \frac{1}{n-k-1}\left[(n-k)MS_{res} - \frac{e_i^2}{1-h_{ii}} \right].
$$

This estimate of σ^2 is used instead of MS_{res} to produce an **externally studentized residual**, usually called ***R*-student** given by

$$
t_i = \frac{e_i}{\sqrt{s_{(i)}^2(1-h_{ii})}},\ i = 1,2,...,n.
$$

In many situations, t_i will differ little from r_i. However, if the ith observation is influential, then $s_{(i)}^2$ can differ significantly from MS_{res} and the $R-$Student statistics will be more sensitive to this point.

Under the usual regression assumption, t follows a t-distribution with $(n-k-1)$ degrees of freedom.

Residual plots

The graphical analysis of residuals is a very effective way to assess the adequacy of a regression model's fit and to check its underlying assumptions. Various types of graphics can be examined for different assumptions, and these graphics are generated by regression software. It is better to plot both the original and scaled residuals. Typically, the studentized residuals are plotted as they have constant variance.

Normal probability plot

The assumption of normality of disturbances is essential for the validity of results in testing of hypotheses, confidence intervals, and prediction intervals. Small departures from normality may not significantly affect the model, but gross nonnormality is more dangerous. Normal probability plots help verify the assumption of normality. If errors are coming from a distribution with thicker and heavier tails than normal, then the least-squares fit may be sensitive to a small set of data. Heavy-tailed error distribution often generates outliers that "pull" the least squares too much in their direction. In such cases, other estimation techniques, such as robust regression methods, should be considered.

The normal probability plot is a plot of the ordered standardised **residuals** versus the so-called normal scores. The normal scores are the cumulative probability

$$P_i = \frac{\left(i - \frac{1}{2}\right)}{n}, \; i = 1, 2, ..., n.$$

If the residuals $e_1, e_2, ..., e_n$ are ordered and ranked in increasing order as

$$e_{[1]} < e_{[2]} < ... < e_{[n]},$$

then the $e_{[i]}'s$ are plotted against P_i and the plot is called a normal probability plot. If the residuals are normally distributed, then the ordered residuals should be approximately the same as the ordered normal scores. So the resulting points should lie approximately on the straight line with an intercept of zero and a slope of one (these are the mean and standard deviation of the standardised residuals).

The rationales behind plotting $e_{[i]}$ against $P_i = \frac{\left(i - \frac{1}{2}\right)}{n}$ is as follows and illustrated in Figure 5:

- Divide the whole unit area under the normal curve into n equal areas.
- We have a sample of size n data sets.
- We might "expect" that one observation lies in each section, so marked out.
- The first section has one point, so the cumulative probability is $P_1 = 1/n$. The second section has one point, so the cumulative probability up to the second section is $P_2 = (1/n) + (1/n) = 2/n$ and so on.
- Then i^{th} ordered residual observation is plotted against the cumulative area to the middle of ith section, which is $\frac{\left(i - \frac{1}{2}\right)}{n}$.
- The factor ½ is used for end correction as all the observations scattered inside the stripe are assumed to be concentrated at the midpoint of the stripe.

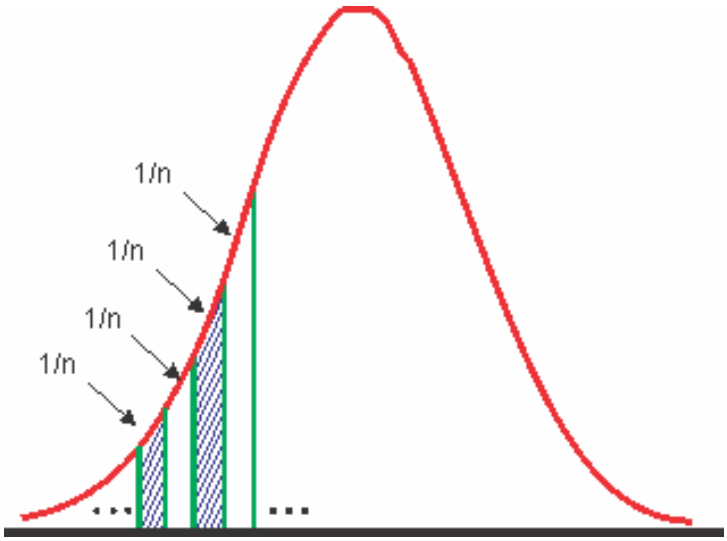


Figure 5: Normal distribution curve

Different software uses different criteria. For example, BMDP uses

$$P_i = \frac{i - \frac{1}{3}}{n + \frac{1}{3}}$$

which produces detrended normal probability plots from which the slope is removed.

Minitab uses $P_i = \dfrac{i - \frac{3}{8}}{n + \frac{1}{4}}$ and converts to a normal score.

Such differences are not crucial in real use.

The straight line is usually determined visually with emphasis on the central values rather than the extremes. A substantial departure from a straight line indicates that the distribution is not normal.

Sometimes, the normal probability plots are constructed by plotting the ranked residuals $e_{[i]}$ against the expected normal value $\Phi^{-1}\left[\dfrac{\left(i - \frac{1}{2}\right)}{n}\right]$ where Φ denotes the standard normal cumulative distribution. This follows from the fact that $E\left[e_{[i]}\right] \approx \Phi^{-1}\left[\dfrac{\left(i - \frac{1}{2}\right)}{n}\right]$.

Various interpretations of the graphic patterns are as follows.

(a) This Figure 6 has an ideal normal probability plot. The points lie approximately on a straight line, indicating that the underlying distribution is normal.

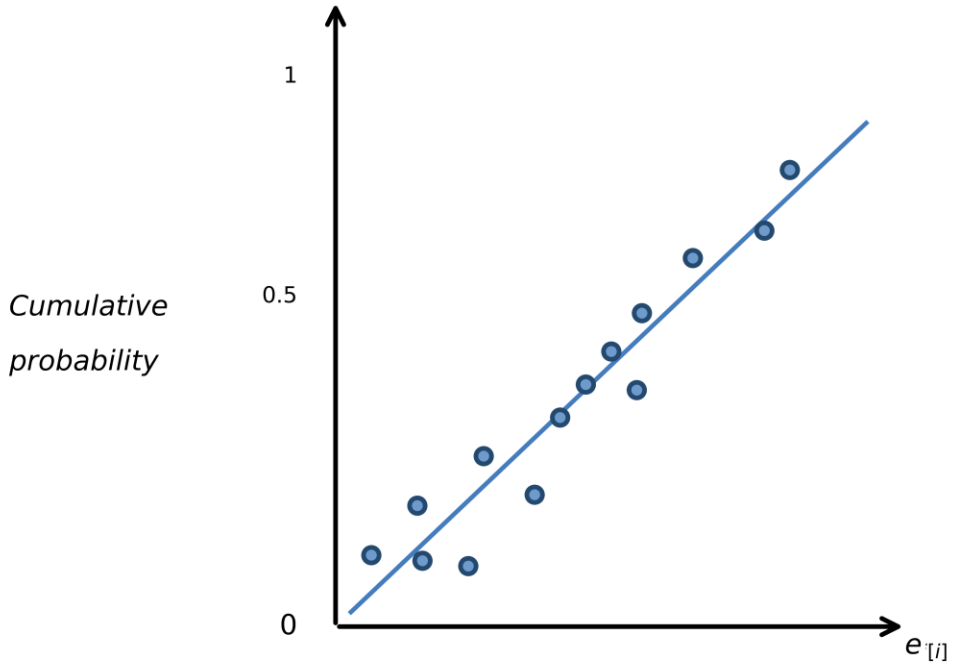


Figure 6: Ideal normal probability plot

(b) This Figure 7 has sharp upward and downward curves at both extremes. This indicates that the underlying distribution is heavy-tailed, i.e., its tails are thicker than those of the normal distribution.

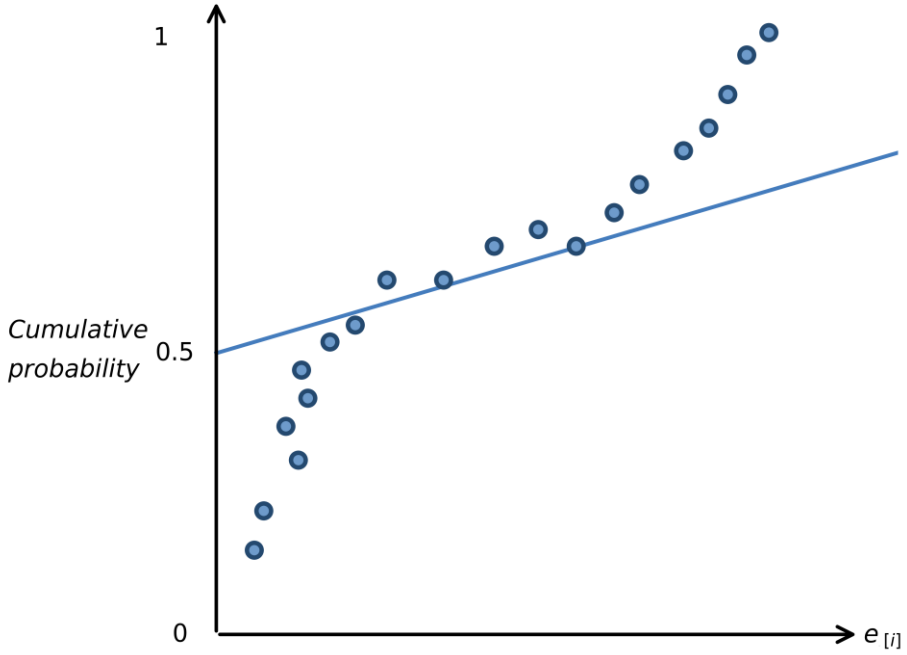


Figure 7: Normal probability plot indicating heavy tailed distribution

(c) This Figure 8 has flattening at the extremes for the curves. This indicates that the underlying distribution is light-tailed, i.e., its tails are thinner than those of the normal distribution.

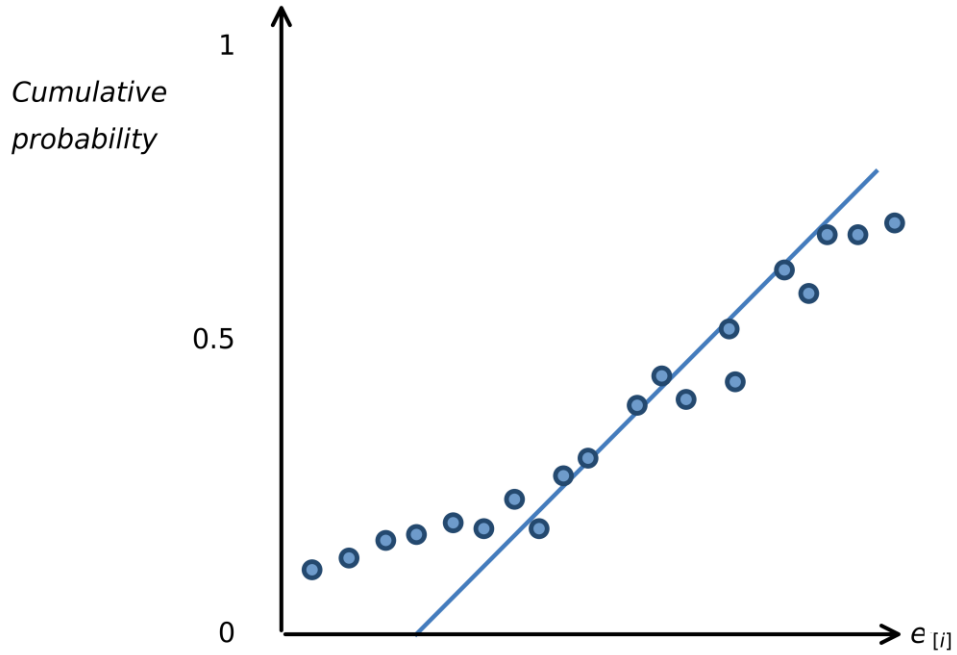


Figure 8: Normal probability plot indicating light tailed distribution

(d) This Figure 9 has a sharp change in the direction of the trend in an upward direction from the middle. This indicates that the underlying distribution is positively skewed.

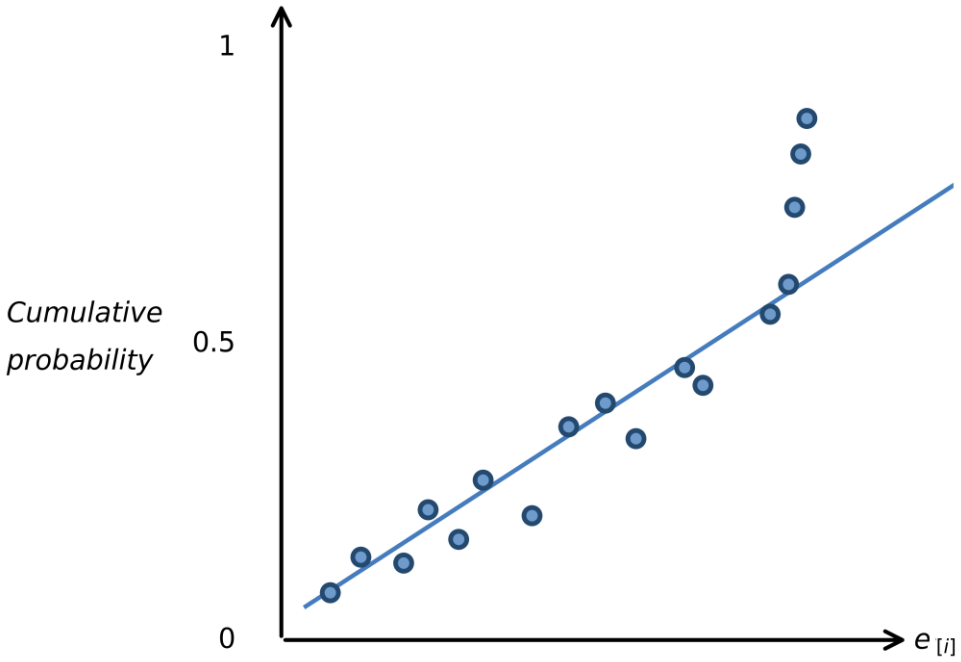


Figure 9: Normal probability plot indicating positively skewed distribution

(e) This Figure 10 has a sharp change in the direction of the trend in the downward direction from the middle. This indicates that the underlying distribution is negatively skewed.

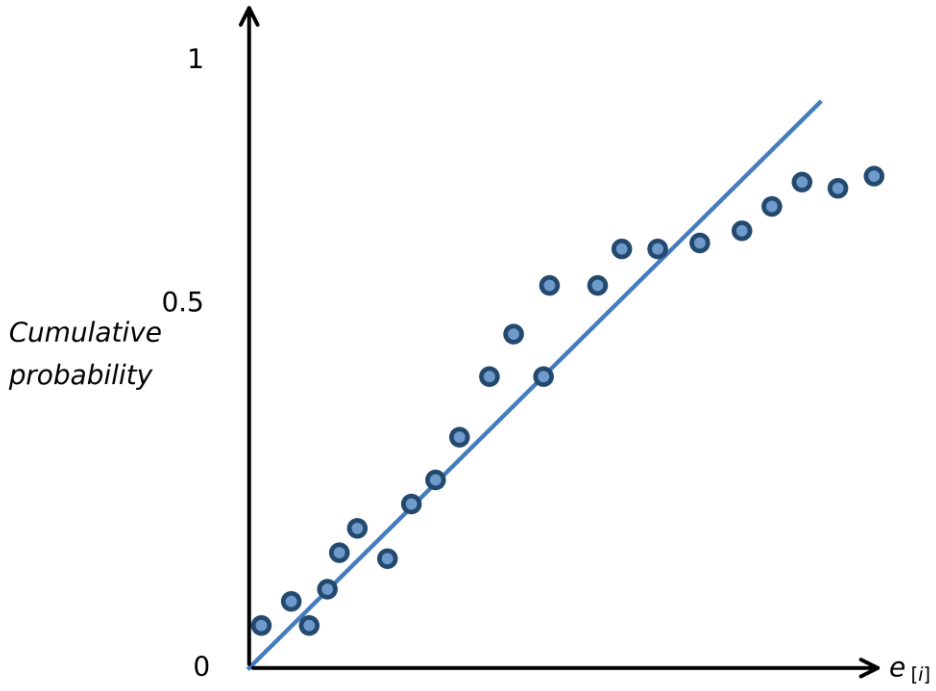


Figure 10: Normal probability plot indicating negatively skewed distribution

Some experience and expertise are required to interpret normal probability plots, because samples from a normal distribution will not plot as a straight line.

- Small sample sizes $(n \le 16)$ often produce normal probability plots that deviate substantially from linearity.
- Larger sample sizes $(n \ge 32)$ produce plots that are much better behaved.
- Usually, about $n = 20$ is required to produce stable and easily interpretable normal probability plots.
- If residuals are not from a random sample, normal probability plots often exhibit no unusual behaviour, even if the disturbances (ε_i) are not normally distributed. Such residuals are often remnants of a parametric estimation process and are linear combinations of the model errors (ε_i).
- Thus, fitting the parameters tends to destroy the evidence of nonnormality in the residuals. Consequently, we cannot rely on normal probability plots to detect departures from normality.
- A commonly seen defect found in normal probability plots is the occurrence of one or two large residuals. Sometimes, this indicates that the corresponding observations are outliers.

Plots of residuals against the fitted value

A plot of residuals (e_i) or any of the scaled residuals $(d_i, r_i$ or $t_i)$ versus the corresponding fitted values $\hat{y}_i$ is helpful in detecting several common types of model inadequacies. The following types of plots of $\hat{y}_i$ versus e_i have particular interpretations:

(a) If the plot is such that the residuals can be contained in a **horizontal band** fashion as in Figure 11 (and the residuals fluctuate more or less in a random manner inside the band), then there are no visible model defects.

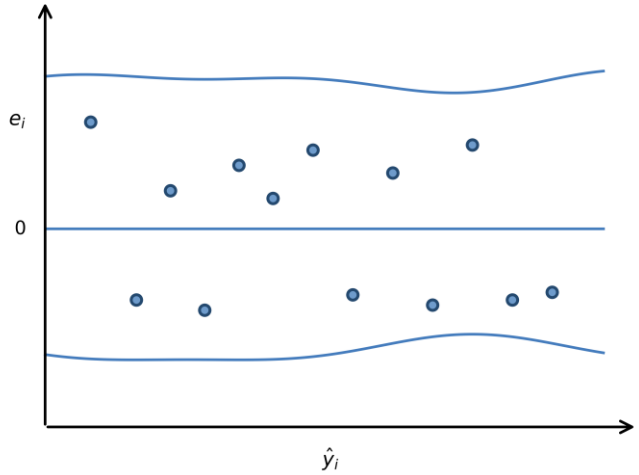


Figure 11: Horizontal band containing residuals

(b) Its plot is such that the residuals can be contained in an **outward opening funnel, as in Figure 12,** then such a pattern indicates that the variance of errors is not constant, but it is an increasing function of y.

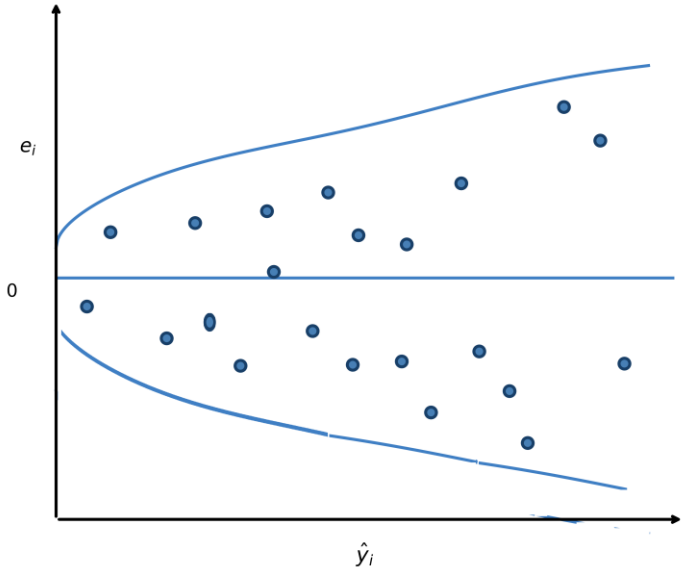


Figure 12: : Outward opening funnel containing residuals

(c) If plots are such that the residuals can be accommodated in an **inward opening funnel,** as in Figure 13, then such a pattern indicates that the variance of errors is not constant, but it is a decreasing function of y.

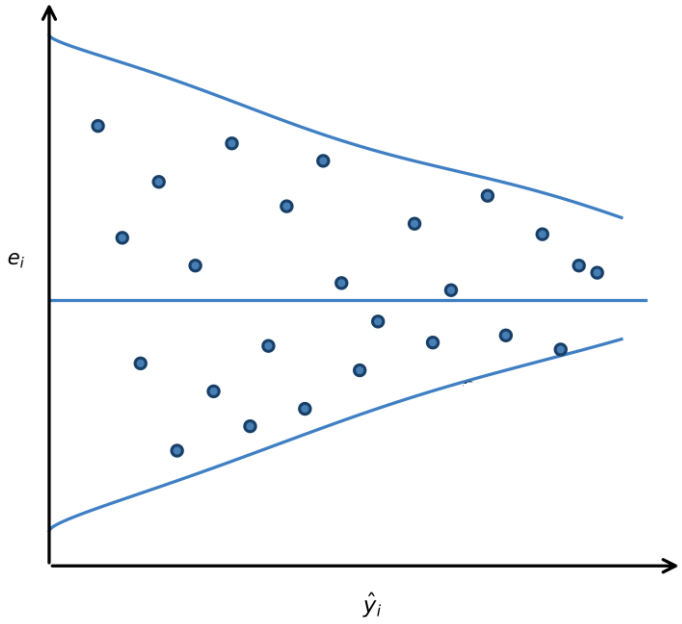


Figure 13: Inward opening funnel containing residuals

(d) If the plot is such that the residuals can be accommodated inside a double bow, then such a pattern indicates that the variance of errors is not constant but y is a proportion between 0 and 1, as in Figure 14. The y then may have a Binomial distribution. The variance of a Binomial proportion near 0.5 is higher than that of near-zero or 1. So the assumed relationship between y and $X's$ is nonlinear.

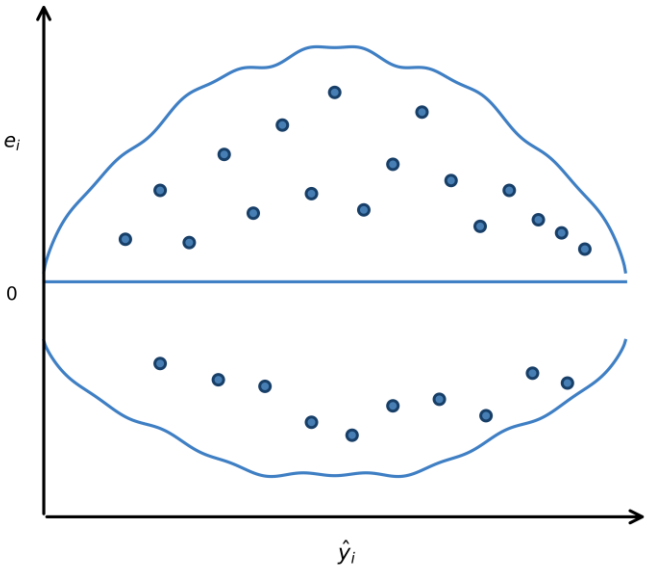


Figure 14: Double bow containing residuals

The usual approach to deal with such inequality of variances is to apply a suitable transformation to either the explanatory variables or the study variable, or use the method of weighted least squares. In practice, transformations on the study variable are generally employed to stabilise the variance.

(e) If the plot is such that the residuals are contained inside a curved plot, as in Figure 15, then it indicates nonlinearity. The assumed relationship between y and $X's$ is non-linear. This could also indicate that additional explanatory variables are needed in the model. For example, a squared error term may be necessary. Transformations on explanatory variables and/or study variable may also be helpful in these cases.

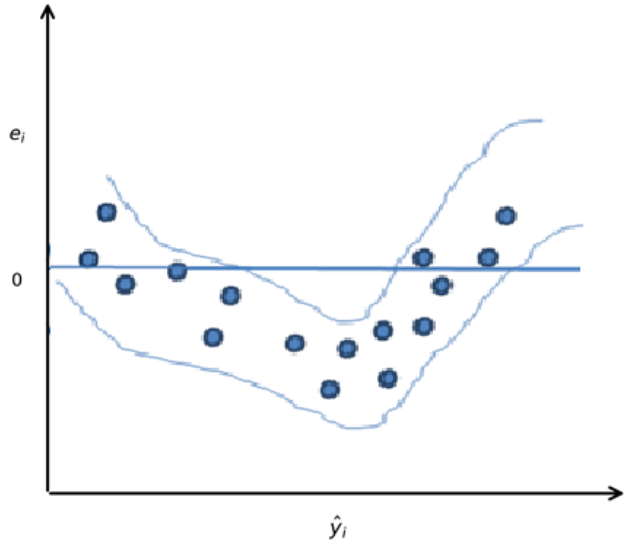


Figure 15: Curved plot containing residuals

Note: A plot of residuals against $\hat{y}_i$ may also reveal one or more unusually large residuals. These points are potential outliers. Large residuals that occur at the extreme $\hat{y}_i$ values could also indicate that either the variance is not constant, or the true relationship between y and X is nonlinear. These possibilities should be investigated before the points are considered outliers.

Plots of residuals against explanatory variable

Plotting of residuals against the corresponding values of each explanatory variable can also be helpful.
We proceed as follows:

- Consider the residual on $Y-$axis and values of the jth explanatory variable $x_{ij}'s,\ (i=1,2,...,n)$ on $X-$axis. This is the same way as we have plotted the residuals against $\hat{y}_i$. In place of $\hat{y}_i's,$ now we consider $x_{ij}'s$.

- Interpretation of the plots is the same as in the case of plots of residuals versus $\hat{y}_i$. This is as follows. If all the residuals are contained in
  - a horizontal band, and the residuals fluctuate more or less in a random fashion within this band, then it is desirable, and there are no visible model defects.
  - an outward opening funnel shape or an inward opening funnel shape, then it indicates that the variance is nonconstant.
  - a double bow pattern or a nonlinear pattern, then it indicates the assumed relationship between y and x_j is not correct. The possibilities like y may be a proportion, a higher-ordered term is X_j (e.g. X_j^2) are needed, or a transformation is required in such a case.

Note 1: In the case of simple linear regression, it is not necessary to plot residuals versus $\hat{y}_i$ and the explanatory variable. The reason is that the fitted values $\hat{y}_i$ are linear combinations of the values of the explanatory variable X_i, so the plots would only differ in the scale for the abscissa ($X-$axis).

Note 2: It is also helpful to plot the residuals against explanatory variables that are **not currently** in the model but which could potentially be included. Any structure in the plot of residuals versus an omitted variable indicates that incorporation of that variable could improve the model.

Note 3: Plotting residuals versus the explanatory variable is not always the most effective way to reveal whether a curvature effect (or a transformation) is required for that variable in the model. Partial regression plots are more effective in investigating the relationship between the study variable and explanatory variables.

Plots of residuals in time sequence

If the time sequence in which the data were collected is known, then the residuals can be plotted against the time order. We proceed as follows:

- Consider the residuals on Y-axis and time order on $X-$axis. This is the same way as we have plotted the residuals against $\hat{y}_i$. In place of $\hat{y}_i$, just use the time order.
- Interpretation of the plots is the same as in the case of plots of residuals versus $\hat{y}_i$. This is as follows.

If all the residuals are contained in

- a horizontal band, and the residuals fluctuate more or less in a random fashion within this band, then it is desirable and indicates that there are no obvious model defects.
- An outward opening funnel shape or an inward opening funnel shape, then it indicates that the variance is not constant but changing with time.
- Double bow pattern or nonlinear pattern, then it indicates that the assumed relationship is nonlinear. In such a case, the linear or quadratic time terms should be added to the model.

The time-series plot of residuals may indicate that errors at one time period are correlated with those at other time periods. The correlation between model errors at different time periods is called **autocorrelation.**
If we have a plot like the following, then it indicates the presence of autocorrelation.

The following type of graphics in Figures 16 and 17 indicates the presence of positive and negative autocorrelation, respectively.

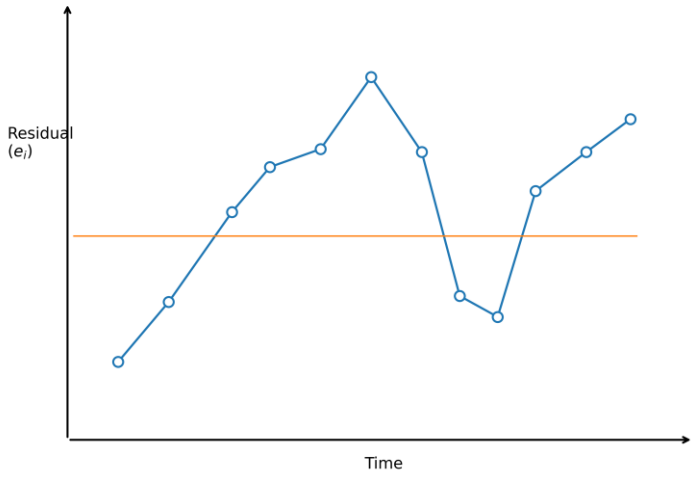


Figure 16: Positive autocorrelation

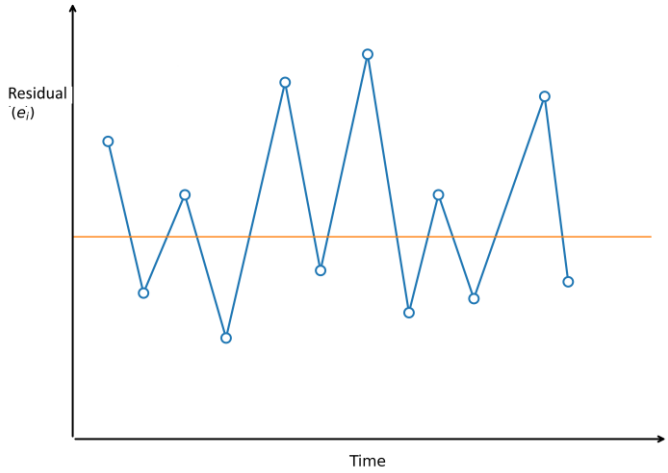


Figure 17: Negative autocorrelation

Methods for detecting autocorrelation and handling time-dependent data are available in time series analysis.

Some measures are discussed further in the module on autocorrelation.

Partial regression and partial residual plots

A partial regression plot (also called an **added-variable plot or adjusted-variable plot**) is a variation of the plot of residuals versus the predictor. It is better to study the marginal relationship of an explanatory variable given the other variables that are in the model. A limitation of the plot of residuals versus an explanatory variable is that it may not completely show the correct or complete marginal effect of an explanatory variable, given the other explanatory variables in the model. The partial regression plot helps evaluate whether the relationship between the study and the explanatory variables is correctly specified. They provide information on the marginal usefulness of a variable not currently in the model.

In a partial regression plot

- Regress y on all the explanatory variables except the jth explanatory variables X_j and obtain the residuals $e\left[y \mid X_{(j)}\right]$, say where $X_{(j)}$ denotes the X-matrix with X_j removed.
- Regress X_j on all other explanatory variables and obtain the residuals $e\left[X_j \mid X_{(j)}\right]$, say
- Plot both these residuals against $e\left[X_j \mid X_{(j)}\right]$.

These plots provide information about the nature of the marginal relationship for j^{th} explanatory variable X_j under consideration.

If X_j enters into the model linearly, then the partial regression plot should show a linear relationship, i.e., the partial residuals will fall along a straight line with a nonzero slope.

See how:

Consider the model

$$\begin{aligned} y &= X\beta + \varepsilon \\ &= X_{(j)}\beta_{(j)} + X_j\beta_j + \varepsilon \end{aligned}$$

then the residual is $e = (I - H)\varepsilon$ where $H = X(X'X)^{-1}X'$ and $H_{(j)} = X_{(j)}(X_{(j)}^{'}, X_{(j)}^{'})^{-1}X_{(j)}^{'}$ is the H-matrix based on $X_{(j)}$. Premultiply $y = X\beta + \varepsilon$ by $(I - H_{(j)})$ and noting that $(I - H_{(j)})X_{(j)} = 0,$ we have

$$\begin{aligned} (I - H_{(j)})y &= (I - H_{(j)})X_{(j)}\beta_{(j)} + (I - H_{(j)})X_j\beta_j + (I - H_{(j)})\varepsilon \\ &= 0 + (I - H_{(j)})X_j\beta_j + (I - H_{(j)})\varepsilon \\ e\left[y \mid X_{(j)}\right] &= e\left[X_j \mid X_{(j)}\right]\beta_j + \varepsilon^* \end{aligned}$$

where $\varepsilon^* = (I - H_{(j)})\varepsilon.$

This suggests that a partial regression plot, which is a plot between $e\left[y \mid X_{(j)}\right]$ and $e\left[X_j \mid X_{(j)}\right]$ (like between y and X) should have a slope β_j. Thus, if X_j linearly enters the regression, the partial regression plot should show a linear relationship passing through the origin.

Example: Consider a model with two explanatory variables X_1 and X_2 as

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon .$$

We want to know about the nature of the marginal relationship for X_1 and also want to know whether the relationship between y and X_1 is correctly specified or not?

To obtain the partial regression plot.

- Regress y on X_2 and obtain the fitted values and residuals

$$\hat{y}_i(X_2) = \hat{\theta}_0 + \hat{\theta}_1 x_{i2}$$
$$e_i(y \mid X_2) = y_i - \hat{y}_i(X_2),\ i = 1, 2, ..., n.$$

- Regress X_1 on X_2 and find the residuals

$$\hat{X}_{i1}(X_2) = \hat{\alpha}_0 + \hat{\alpha}_1 x_{i2}$$
$$e_i(X_1 \mid X_2) = x_{i1} - \hat{X}_{i1}(X_2),\ i = 1, 2, ..., n.$$

- Plot $e_i(y \mid X_2)$ against the X_1 residuals $e_i(X_1 \mid X_2)$.

Case 1. If X_1 enters into the model linearly, then the plot will look as in Figure 18. Then the slope of this line is the regression coefficient of X_1 in the multiple linear regression model.

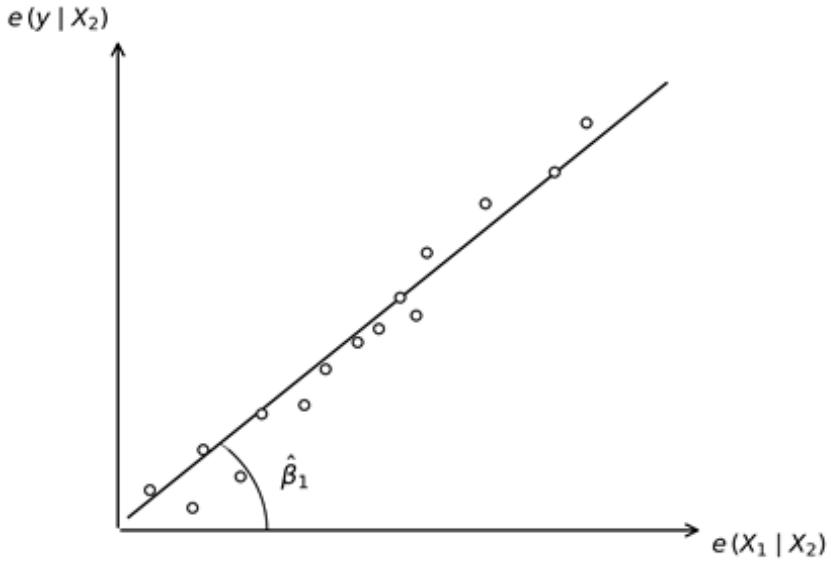


Figure 18: Linear partial regression plot

Case 2: If the partial regression plot shows a curvilinear band, as in Figure 19, then higher-order terms in X_1 or a transformation X_1 (or equivalently, in general , X_j) may be helpful.

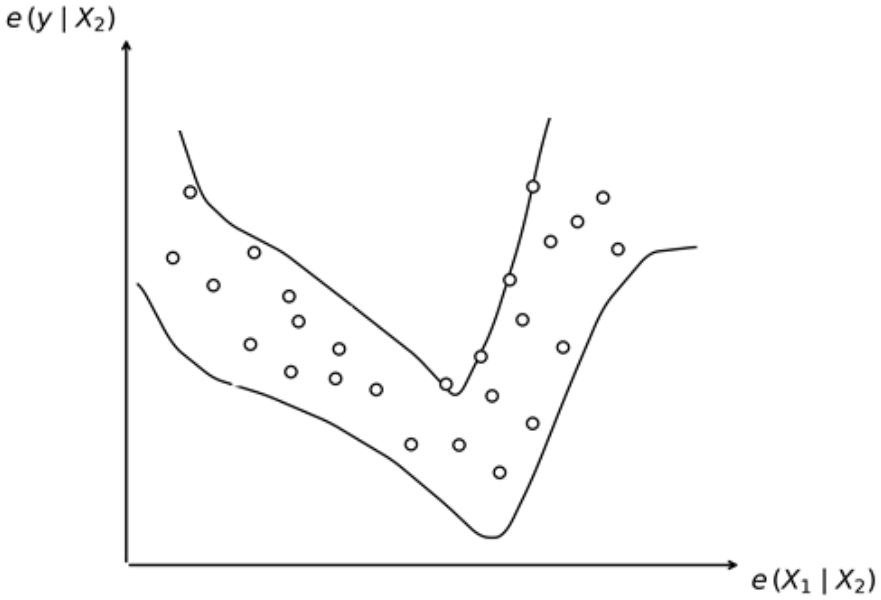


Figure 19: Curvilinear partial regression plot

Case 3: If X_1 is a candidate variable which is considered for inclusion in the model, as in Figure 20, then a horizontal band on the regression plot indicates that there is no additional useful information for predicting y. In general, if X_j is a candidate variable that is considered for inclusion in the model, then a horizontal band on the regression plot indicates that there is no additional useful information in X_j for predicting y. This indicates that β_j is nearly zero.

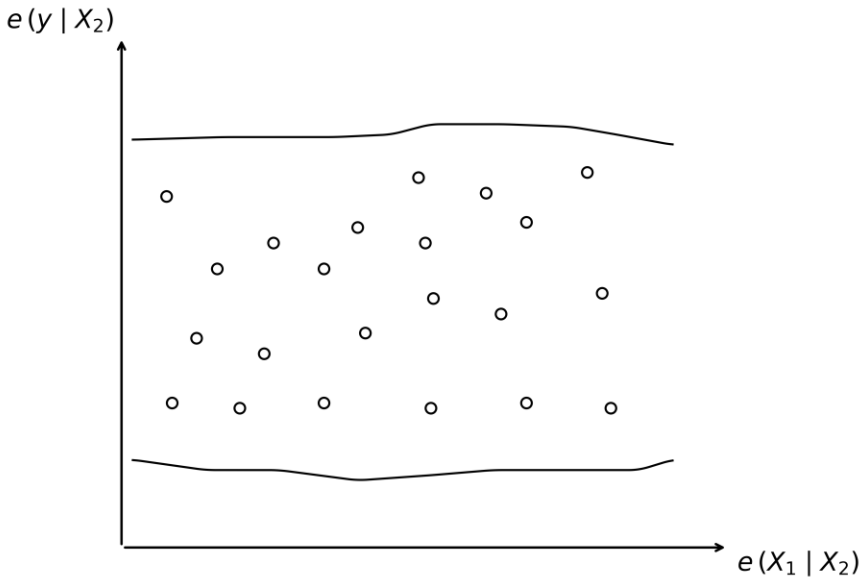


Figure 20: Horizontal band in partial regression plot

Some comments on partial regression plots

1. Partial regression plots need to be used with caution as they only suggest a possible relationship between the study and explanatory variables. The plots may not reveal the proper form of the relationships among variables already in the model that are incorrectly specified.

   Some alternative forms of the relationship between the study and explanatory variables should also be examined using several transformations.

   Residual plots for these models should also be examined to identify the best relationship or transformation.
2. Partial regression plots will not, in general, detect the interaction effect among the regressors.
3 Partial regression plots are affected by the presence of an exact relationship among explanatory variables (multicollinearity), and the information about the relationship between the study and the explanatory variables may be incorrect.
4 In such cases, it is better to construct a scatter plot of explanatory variables like X_i verus X_j. If they are highly correlated, multicollinearity is introduced, and properties of estimators like ordinary least squares of regression coefficients are disturbed.

Partial residual

A residual plot closely related to the partial regression plot in the partial residual plot. It is designed to show the relationship between the study and explanatory variables. Partial residuals are those residuals that have not been adjusted for particular regressor.

Suppose the model has k explanatory variable, and we are interested in the jth explanatory variable X_j. Then $X = (X_{(j)}, X_j)$ where $X_{(j)}$ is the $X-$matrix with X_j removed.

The model is

$$\begin{aligned} y &= X\beta + \varepsilon \\ &= X_{(j)}\beta_{(j)} + X_j\beta_j + \varepsilon \end{aligned}$$

where $\beta_{(j)}$ is the vector of all $\beta_1, \beta_2, ..., \beta_k$ except β_j. The fitted model is

$$\hat{y} = X_{(j)}\hat{\beta}_{(j)} + X_j\hat{\beta}_j + e$$

or $\hat{y} - X_{(j)}\hat{\beta}_{(j)} = X_j\hat{\beta}_j + e$

where e is the residual based on all k explanatory variables.

Then the partial residual for X_j $(j = 1,2,..,k)$ is given by

$$\hat{y} - X_{(j)}\hat{\beta}_{(j)} = X_j\hat{\beta}_j + e$$

or $e(y \mid X_{(j)}) = e + \hat{\beta}_j X_j, i = 1,2,...,n.$

Partial residual plots

A residual plot closely related to the partial regression plot in the partial residual plot. It is designed to show the relationship between the study and explanatory variables.

Suppose the model has k explanatory variables $X_1, X_2, ..., X_k$. The partial residuals for X_j are defined as

$$e(y \mid X_{(j)}) = e_i + \hat{\beta}_j x_{ij},\ i = 1,2,...,n$$

where e_i are the residuals from the model containing all the k explanatory variables and $\hat{\beta}_j$ is the estimate of the jth regression coefficient.

When $e(y \mid X_{(j)})$ are plotted against x_{ij}, the resulting display has a slope $\hat{\beta}_j$. The interpretation of the partial residual plot is very similar to that of the partial regression plot.

Statistical tests on residuals

We may apply statistical tests to the residuals to obtain quantitative measures of some model inadequacies. They are not widely used. In many applications, residual plots are more informative than the corresponding tests. However, some residual plots do require some skill and experience to interpret. In such cases, the statistical tests may prove useful.

The PRESS statistic

The PRESS residuals are defined as

$$e_{(i)} = y_i - \hat{y}_{(i)},\ i = 1,2,...,n$$

where $\hat{y}_{(i)}$ is the predicted value of the ith observed study variable based on a model fit to the remaining $(n-1)$ points. Large residuals are useful for identifying observations where the model does not fit well or is likely to provide poor predictions for future values.

The prediction sum of squares is defined as the sum of squared PRESS residuals and is called the PRESS statistic as

$$PRESS = \sum_{i=1}^{n}\left[y_i - \hat{y}_{(i)}\right]^2 = \sum_{i=1}^{n}\left[\frac{e_i}{1-h_{ii}}\right]^2$$

The PRESS statistic is a measure of how well a regression model will perform in predicting new data. So this is also a measure of model quality. A model with a low PRESS value is desirable. This can also be used for comparing regression models.

R^2 for prediction based on PRESS

The PRESS statistic can be used to compute an R^2-like statistic for prediction, say

$$R^2_{\text{prediction}} = 1 - \frac{PRESS}{SS_T}$$

where SS_T is the total sum of squares. This statistic indicates the predictive capability of the regression model. For example, if $R^2 = 0.89$, then it indicates that the model is expected to explain about 89% of the variability in predicting new observations.

Detection and treatment of outliers

- An outlier is an extreme observation.
- Residuals that are considerably larger in absolute value than the others, say, 3 or 4 times the standard deviation from the mean, indicate potential outliers in y-space. This idea is based on 3-sigma or 4-sigma limits.
- Depending on their location, outliers can have moderate to severe effects on the regression model.
- Outliers may indicate a model failure for these points.
- Residual plots against $\hat{y}_i$ and normal probability plots help in identifying outliers. Examination of scaled residuals, e.g., studentized and *R*-student residuals, is more helpful as they have mean zero and variance one.
- Outliers can also occur in explanatory variables in X-space. They can also affect the regression results.
- Sometimes outliers are "bad" values occurring as a result of unusual but explainable events. For example, faulty measurements, incorrect data recording, failure of the measuring instrument, etc.

- Bad values need to be discarded, but should have strong nonstatistical evidence that the outlier is a bad value before it is discarded. Discarding bad values is desirable because least-squares pulls the fitted equation toward the outlier.
- Sometimes an outlier is an unusual but perfectly plausible observation. If such observations are deleted, then it may give a false impression of improvement in the fit of the equation.
- Sometimes the outlier is more critical than the rest of the data because it may control many key model properties.
- The effect of outliers on the regression model may be checked by dropping these points and refitting the regression equation.
- The value of t-statistic, F-statistic, R^2 and residual mean square may be sensitive to outliers.

An outlier test based on *R*-student

A common way to model outliers is the mean-shift outlier model.

Suppose we fit a model

$$y = X\beta + \varepsilon$$

when the true model is

$$y = X\beta + \delta + \varepsilon$$

where δ is a $n \times 1$ vector of zeros except for the uth observation, which has a value δ_u. Thus

$$\delta = (0, 0, ..., 0, \delta_u, 0, ..., 0)$$

Assume $\varepsilon \sim N(0, \sigma^2 I)$ for both the models we fit. Our objective is to find an appropriate statistic for testing $H_0 : \delta_u = 0$ verus $H_0 : \delta_u \neq 0$. This procedure assumes that we are specifically interested in u^{th} observation, i.e., that we have a priori information that the uth observation may be an outlier.

First, we find an appropriate estimate of δ_u. Consider u^{th} residual as its estimate. The $n \times 1$ residual vector is

$$e = [I - H]y = \left[I - X(X'X)^{-1}X'\right]y.$$

Then

$$\begin{aligned} E(e) &= \bar{H}y \\ &= \bar{H}E(y) \\ &= \bar{H}(X\beta + \delta) \\ &= \bar{H}X\beta + \bar{H}\delta \\ &= 0 + [I - H]\delta \\ &= \left[I - X(X'X)^{-1}X \right]\delta. \end{aligned}$$

Thus $E(e_u) = (1 - h_{uu})\delta_u$

$$\Rightarrow \hat{\delta}_u = \frac{e_u}{1 - h_{uu}}$$

is an unbiased estimator of δ_u where h_{uu} is the uth diagonal element of H.

It may be observed that $\hat{\delta}_u$ is simply the uth PRESS residual. Further, the covariance matrix of e is

$$\begin{aligned} V(e) &= V\left[(I - H)y\right] \\ &= (I - H)V(y)(I - H) \\ &= \sigma^2(I - H) \\ Var(e_u) &= (1 - h_{uu})\sigma^2. \end{aligned}$$

So

$$\begin{aligned} Var(\hat{\delta}_u) &= Var\left(\frac{e_u}{1 - h_{uu}} \right) \\ &= \frac{(1 - h_{uu})\sigma^2}{(1 - h_{uu})^2} \\ &= \frac{\sigma^2}{1 - h_{uu}}. \end{aligned}$$

Also e is a linear combination of normally distributed y. So e is also normally distributed. Thus $\hat{\delta}_u$ is also normally distributed.

Consequently, under $H_0 : \delta_u = 0$,

$$\frac{\left(\frac{e_u}{1 - h_{uu}} \right)}{\left(\frac{\sigma}{\sqrt{1 - h_{uu}}} \right)} = \frac{e_u}{\sigma\sqrt{1 - h_{uu}}} \sim N(0,1).$$

The quantity $\frac{e_u}{\sigma\sqrt{1-h_{uu}}}$ is simply an example of a studentized residual. Since σ^2 is unknown and $\frac{MS_{res}}{\sigma^2}$ is a Chi-square random variable, so a candidate test statistic is

$$\frac{e_u}{\sqrt{MS_{res}(1-h_{uu})}}$$

which follows a *t*-distribution if $e=[I-H]y$ and $SS_{res}=y'(I-H)y$ are independent.

Since

$$[I-H]\sigma^2 I[I-H]=\sigma^2(I-H)\neq 0,$$

so e and SS_{res} are not actually independent.

We have already developed $s^2_{(u)}$ which is related to the residual mean square in a regression model with u^{th} observation withheld given by

$$s^2_{(u)}=\frac{1}{n-k-1}\left[(n-k)MS_{res}-\frac{e_u^2}{1-h_{uu}}\right].$$

This estimate of σ^2 is independent of e_u by the basic independence assumption on random errors. So σ^2 can be replaced by $s^2_{(u)}$ and an appropriate test statistic for the mean shift outlier model is

$$\frac{e_u}{s_{(u)}\sqrt{1-h_{uu}}}$$

which is the externally studentized residual or R-student.

Under $H_0:\delta_u=0,\quad \frac{e_u}{s_{(u)}\sqrt{1-h_{uu}}}\sim t(n-k-1)$

and under $H_0:\delta_u\neq 0,\quad \frac{e_u}{s_{(u)}\sqrt{1-h_{uu}}}\sim$ noncentral $t[(n-k-1,\gamma]$

with noncentrality parameter

$$\gamma=\frac{\delta_u}{\sigma/(\sqrt{1-h_{uu}})}=\frac{\delta_i\sqrt{1-h_{uu}}}{\sigma}.$$

Note that the power of this test depends on h_{uu}. If we fit an intercept to our model, then $\frac{1}{n}\leq h_{uu}\leq 1$. So, maximum power occurs when $h_{uu}=\frac{1}{n}$, i.e., at the center of the data cloud in terms of the $X's$. As $h_{uu}\to 1$, the power goes to 0. In other words, this test has less ability to detect outliers at the high-leverage data points (Note that the concept of leverage points is discussed in later sections).

Test for lack of fit of a regression model

This test for lack of fit of a regression model is based on the assumptions of normality, independence, and constant variance, which are satisfied. Only the first order or straight-line character of the relationship is in doubt.

For example, the data in the following scatter plot in Figure 21 indicates that a straight-line fit is not very satisfactory.

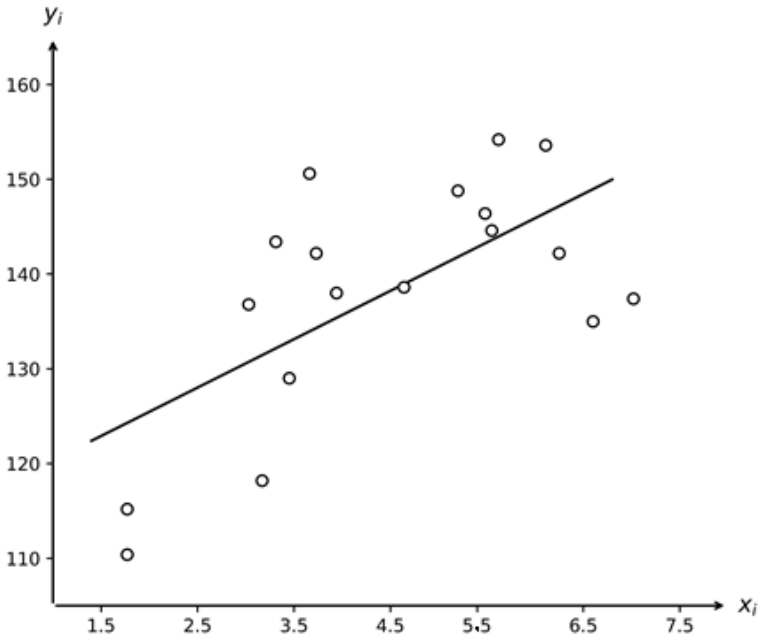


Figure 21: Unsatisfactory straight line fit

The test procedure determines whether systematic curvature is present. The test requires replicating observations on y for at least one level of x , and they should be true replications and not just duplicate readings or measurements of y.

The true replications consist of running n_i separate experiments at $x = x_i$ and observe y. It is not just running a single experiment at $x = x_i$ and measuring y n_i times in which the information is only on the variability of the method of measuring y is obtained. These replicated observations are used to get a model-independent estimate of σ^2.

Suppose we have n_i observations on y at the ith level of $x_i, i = 1,2,...,m.$ Let y_{ij} be the jth observation on y at x_i, $i = 1,2,...,m, j = 1,2,...,n_i;$ $n = \sum_{i=1}^{m} n_i$ is the total number of observations.

Consider the model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i .$$

Let $\bar{y}_i$ be the mean of n_i observations on x_i. Then the $(i,j)^{th}$ residual is

$$(y_{ij} - \hat{y}_i) = (y_{ij} - \bar{y}_i) + (\bar{y}_i - \hat{y}_i)$$

$$\sum_{i=1}^{m}\sum_{j=1}^{n_i}(y_{ij} - \hat{y}_i)^2 = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_i)^2 + \sum_{i=1}^{m} n_i(\bar{y}_i - \hat{y}_i)^2 \text{ (obtained by squaring and summing over } i \text{ and } j)$$

$$\downarrow \qquad\qquad \downarrow \qquad\qquad \downarrow$$

$$SS_{res} \quad = \quad SS_{PE} \quad + \quad SS_{LOF}$$

$$\downarrow \qquad\qquad \downarrow \qquad\qquad \downarrow$$

Residual sum of squares	Sum of squares due to pure error	Sum of squares due to lack of fit
	↓	↓
	Measures pure error	Measures lack of fit

If the assumption of constant variance is satisfied, then SS_{PE} is a **model-independent measure of pure error** because only the variability of $y's$ at each x level is used to compute SS_{PE}.

Since there are $(n_i - 1)$ degrees of freedom for pure error at each level of x_i, the number of degrees of freedom associated with SS_{PE} is $\sum_{i=1}^{m}(n_i - 1) = n - m$. SS_{LOF} is a weighted sum of squared deviations between $\bar{y}_i$ at each level of x and the corresponding fitted value.

If $\hat{y}_i$ are close to $\bar{y}_i$, then there is a strong indication that the regression function is linear.

If $\hat{y}_i$ deviate considerably from $\bar{y}_i$ then it is likely that the regression function is not linear. The degrees of freedom associated with SS_{LOF} is $m-2$ because there are m levels of x and two degrees of freedom are lost because two parameters must be estimated to obtain $\bar{y}_i$.

Computationally,

$$SS_{LOF} = SS_{res} - SS_{PE} .$$

The test statistic for lack of fit is

$$F_0 = \frac{SS_{LOF}/(m-2)}{SS_{PE}/(n-m)}$$

$$= \frac{MS_{LOF}}{MS_{PE}}$$

$$E(MS_{LOF}) = \sigma^2 + \frac{\sum_{i=1}^{n} n_i \left[E(y_i) - \beta_0 - \beta_1 x_i\right]^2}{(m-2)}.$$

If true regression is linear, then $E(y_i) = \beta_0 + \beta_1 x_i$ and $E(MS_{LOF}) = \sigma^2$.

If true regression is nonlinear, then $E(y_i) \neq \beta_0 + \beta_1 x_i$ and $E(MS_{LOF}) > \sigma^2$.

If the true regression function is linear, then

$$F_0 \sim F(m-2, n-m).$$

So to test for lack of fit, compute F_0 and conclude that the regression function is not linear if $F_0 > F_\alpha(m-2, n-m)$ at α level of significance.

If we conclude that the regression function is not linear, then the tentative model must be abandoned, and we attempt to find a more appropriate model.

If $F_0 < F_\alpha(m-2, n-m)$ then there is no strong evidence of a lack of fit. They MS_{PE} and MS_{LOF} are often combined to estimate σ^2.

If F ratio for lack of fit is not significant and $H_0 : \beta_1 = 0$ is rejected, then this does not guarantee that the model will be satisfactory for prediction. It is suggested that the F-ratio must be at least four or five times the $F_\alpha(m-2, n-m)$ if the regression model is to be useful for prediction.

A simple measure of potential prediction performance is found by comparing the range of fitted values, i.e., $\left(\hat{y}_{\max} - \hat{y}_{\min}\right)$ to their average standard error. Regardless of the term of the model, the average variance of the fitted values is

$$\overline{Var(\hat{y})} = \frac{1}{n}\sum_{i=1}^{n} Var(\hat{y}_i) = \frac{k\sigma^2}{n}$$

where k is the number of parameters in the model.

In general, the model is not likely to be a satisfactory predictor unless the range of $\hat{y}_i$ is large relative to the estimated standard error $\sqrt{\frac{k\hat{\sigma}^2}{n}}$ where $\hat{\sigma}^2$ is a model-independent estimate of error variance.

Estimation of pure error from near-neighbours

In a test of lack of fit

$$SS_{res} = SS_{PE} + SS_{LOF}$$

SS_{PE} is computed using responses at repeat observations at some level of x. This is a model-independent estimate of σ^2.

This general principle applies to any regression model.

Calculation of SS_{PE} requires repeated observations on the response y at the same set of levels on the explanatory variables $x_1, x_2 ..., x_{k,}$ i.e., some of the rows of X-matrix must be the same.

In practice, repeated observations do not often occur in multiple regression, and the procedure of lack of fit is not often useful.

A method to obtain a model-independent estimate of error when there are no exact repeat points are the procedure that searches for those points in x-space that are near neighbors.

This is the set of observations that have been taken with near-identical levels of $x_1, x_2, ..., x_k$. The response y_i from such near-neighbours can be considered as repeat points and used to obtain an estimate of pure error.

As a measure of the distance between any two points, $x_{i1}, x_{i2}, ..., x_{ik}$ and $x_{i'1}, x_{i'2}, ..., x_{i'k}$, use the weighted sum of squared distances (WSSD)

$$D_{ii'}^2 = \sum_{j=1}^{k} \left[\frac{\hat{\beta}_j (x_{ij} - x_{i'j})}{\sqrt{MS_{res}}} \right]^2$$

The pairs of points with small values of $D_{ii'}^2$ are "near neighbours", i.e., they are relatively close together in x-space. Pairs of points for which $D_{ii'}^2$ is large (e.g., $D_{ii'}^2 >> 1$) are widely separated in x-space. The residuals at two points with a small value of $D_{ii'}^2$ can be used to obtain an estimate of pure error.

The estimate is obtained from the range of residuals at the points i and i', say

$$E_i = |e_i - e_{i'}|.$$

There is a relationship between the range of a sample from a normal population and the population standard deviation. For example, for a sample of size 2, this relationship is

$$\hat{\sigma} = \frac{E}{1.128} = 0.886E .$$

The quantity $\hat{\sigma}$ so obtained is an estimate of the standard deviation of pure error.

An efficient algorithm may be used to compute this estimate as follows:

- First arrange the data points $x_{ii},...,x_{ik}$ in order of increasing $\hat{y}_i$.
- Note that points with different values of $\hat{y}_i$ cannot be near neighbours but those with similar values of $\hat{y}_i$ could be neighbours (or they could be near the same contour of constant $\hat{y}$ but for apart in some x-coordinates).

Then

1. Compute the values of $D^2_{ii'}$ for all $(n-1)$ pairs of points with adjacent values of $\hat{y}$. Repeat this calculation for the pairs of points separated by one, two and three intermediate $\hat{y}$ values. This will produce $(4n-10)$ values of $D^2_{ii'}$.
2. Arrange the $(4n-10)$ values of $D^2_{ii'}$ found is step 1. Let $E_u, u = 1,2,...,(4n-10)$ be the range of the residuals at these points.
3. For the first m values of E_u, calculate an estimate of the standard deviation of pure error as

$$\hat{\sigma} = \frac{0.886}{m}\sum_{u=1}^{m} E_u .$$

Note that $\hat{\sigma}$ is based on the average range of the residuals associated with the m smallest values of $D^2_{ii'}, m$ must be chosen after inspecting the values of $D^2_{ii'}$. One should not include the values of E_u in the calculation for which the weighted sum of squared distances is too large.

Transformation and Weighting to Correct Model Inadequacies

Graphical methods help detect violations of basic assumptions in regression analysis. Now we consider methods and procedures for building models that use data transformation when assumptions are violated.

Variance stabilizing transformations

In regression analysis, it is assumed that the variance of disturbances is constant, i.e., $Var(\varepsilon_i) = \sigma^2,\ i = 1,2,...,n.$ Suppose this assumption is violated. A common reason for such isolation is that the study variable follows a probability distribution whose variance is a function of the mean.

Some commonly used variance-stabilizing transformations in the order of their strength are as follows:

Relation of σ^2 to $E(y)$	**Transformation**
$\sigma^2 \propto$ constant	$y^* = y$ **(**no transformation**)**
$\sigma^2 \propto E(y)$	$y^* = \sqrt{y}$ (Poisson data)
$\sigma^2 \propto E(y)[1-E(y)]$	$y^* = \sin^{-1}(\sqrt{y})$ **(**Binomial proportion $0 \le y_i \le 1$**)**
$\sigma^2 \propto [E(y)]^2$	$y^* = \ln(y)$
$\sigma^2 \propto [E(y)]^3$	$y^* = 1/\sqrt{y}$
$\sigma^2 \propto [E(y)]^4$	$y^* = \frac{1}{y}$

After making a suitable transformation, use y^* as a study variable in the respective case.

The strength of a transformation depends on the degree of curvature present in the curve between the study and the explanatory variable. The transformation mentioned here ranges from relatively mild to relatively strong. The square root transformation is relatively mild, and the reciprocal transformation is relatively strong. The square root transformation is relatively mild, and the reciprocal transformation is relatively strong.

In general, a mild transformation is applied when the minimum and maximum values do not range much (e.g. $y_{\max} / y_{\min} < 2,3)$ and such a transformation has little effect on the curvature. On the other hand, when the minimum and maximum vary greatly, a strong transformation is needed to have a substantial impact on the analysis.

Analytical methods for selecting a transformation on the study variable

The Box-Cox method

Suppose the normality and/or constant variance of the study variable y can be corrected through a power transformation on y. This means y is to be transformed as y^λ where λ is the parameter to be determined. For example, if $\lambda = 0.5,$ then the transformation is the square root and $\sqrt{y}$ is used as a study variable in place of y.

Now the linear regression model has parameters β, σ^2 and λ. The Box-Cox method tells how to estimate simultaneously the λ and parameters of the model using the method of maximum likelihood.

Note that as λ approaches zero, y^λ approaches to 1. So there is a problem at $\lambda = 0$ because this makes all the observations y to be one. It is meaningless for all observations on the study variable to be constant. So there is a discontinuity at $\lambda = 0$. One approach to solve this difficulty is to use $\frac{y^\lambda - 1}{\lambda}$ as a study variable.

Note that as $\lambda \to 0, \frac{y^\lambda - 1}{\lambda} \to \ln y$. So, a possible solution is to use the transformed study variable as

$$y^{(\lambda)} = V = \begin{cases} \dfrac{y^\lambda - 1}{\lambda y_*^{\lambda - 1}} & \text{for } \lambda \neq 0 \\ y_* \ln y & \text{for } \lambda = 0 \end{cases}$$

where y_* is the geometric mean of y_i's as $y_* = (y_1 y_2 ... y_n)^{1/n}$, which is constant.

When V is applied to each y_i, we get $V = (V_1, V_2, ..., V_n)'$ as a vector of observations on the transformed study variable, and we use it to fit a linear model

$$V = X\beta + \varepsilon$$

using the least squares or maximum likelihood method.

To find the appropriate family, consider

$$y^{(\lambda)} = V = X\beta + \varepsilon$$

where $y^{(\lambda)} = \frac{y^\lambda - 1}{\lambda y_*^{\lambda - 1}}, \varepsilon \sim N(0, \sigma^2 I).$

Applying the method of maximum likelihood to the likelihood function for $y^{(\lambda)}$, the log-likelihood function $\ln L\left[y^{(\lambda)}\right]$ gives

$$L(\lambda) = -\frac{n}{2}\ln\hat{\sigma}^2 = -\frac{n}{2}\ln\left[SS_{res}(\lambda)\right]$$

where $SS_{res}(\lambda)$ is the sum of squares due to residuals, which is a function of λ. Now maximize $L(\lambda)$ with respect to λ. It is difficult to obtain any closed form of the estimator of λ. So we maximize it numerically.

The function $-\frac{n}{2}\ln\left[SS_{res}(\lambda)\right]$ is called as the **Box-Cox objective function.**

Computational procedure

The maximum- likelihood estimate of λ corresponds to the value of λ for which residual sum of squares from the fitted model $SS_{res}(\lambda)$ is a minimum. To determine such λ, we proceed computationally as follows:

- Fit $y^{(\lambda)}$ for various values of λ. For example, start with values in (-1, 1), then take the values in (-2, 2) and so on. Take about 15 to 20 values of λ which are expected to be sufficient for the estimation of optimum value.
- Plot $SS_{res}(\lambda)$ versus λ.
- Find the value of λ which minimizes $SS_{res}(\lambda)$ from the graph.
- A second iteration can be performed using a finer mesh of values of the desired.

Note that the value of λ can not be selected by directly comparing the residual sum of squares from the regression of y^{λ} on x because for each λ, the residual sum of squares is measured on a different scale.
It is better to use simple values of λ. For example, the practical difference between $\lambda = 0.5$ and $\lambda = 0.58$ is likely to be small but $\lambda = 0.5$ is much easier to interpret.

Once λ is selected, then use

- y^{λ} as a study variable if $\lambda \neq 0$
- $\ln y$ as a study variable if $\lambda = 0$.

It is entirely acceptable to use $y^{(\lambda)}$ as a response to the final model. This model will have a scale difference and an origin shift in comparison to the model using y^{λ} (or $\ln y$) as the response.

An approximate confidence interval for λ

We can find an approximate confidence interval for the transformation parameter λ. This interval helps in selecting the final value of λ. For example, if $\hat{\lambda} = 0.58$ is the value of λ which is minimizing the sum of squares due to the residual. But if $\lambda = 0.5$ is within the confidence interval, then one may use the square-root transformation because it is easier to explain. Furthermore, if $\lambda = 1$ is in the confidence interval, then it may be concluded that no transformation is necessary.

In applying the method of maximum likelihood to the regression model, we are essentially maximizing

$$L(\lambda) = -\frac{n}{2}\ln\left[SS_{res}(\lambda)\right]$$

or equivalently, we are minimizing $SS_{res}(\lambda)$.

An approximate $100(1-\alpha)\%$ confidence interval for λ consists of those values of λ that satisfy

$$L(\hat{\lambda}) - L(\lambda) \leq \frac{\chi^2_\alpha(1)}{2}$$

where $\chi^2_\alpha(1)$ is the upper $\alpha\%$ point of the Chi-square distribution with one degree of freedom.

The approximate confidence interval is constructed using the following steps:

- Draw a plot of $L(\lambda)$ versus λ as in Figure 22.
- Draw a horizontal line at the height

  $$L(\hat{\lambda}) - \frac{\chi^2_\alpha(1)}{2}$$

  on the vertical scale.
- This line would cut the $L(\lambda)$ at two points.
- The location of these two points on the λ-axis defines the two endpoints of the approximate confidence interval.
- If the sum of squares due to residuals is minimized and $SS_{res}(\lambda)$ versus λ is plotted, then the line must be plotted at the height

  $$SS^* = SS_{res}(\hat{\lambda})\exp\left[\frac{\chi^2_\alpha(1)}{n}\right]$$

  where $\hat{\lambda}$ is the value of λ which minimizes the sum of squares due to residuals.

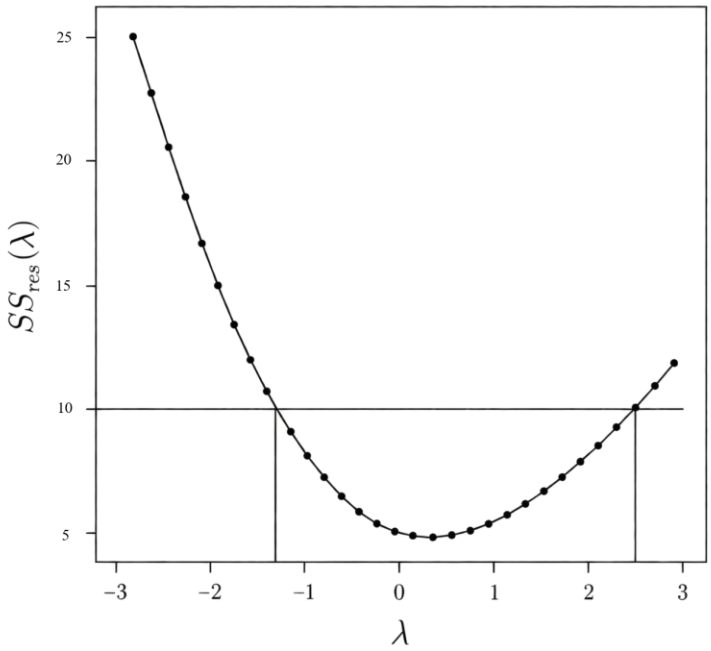


Figure 22: Plot of $SS_{res}(\lambda)$ versus λ

Using the expansion of the exponential function as

$$\exp(t) = 1 + t + \frac{t^2}{2!} + \ldots$$
$$\approx 1 + t,$$

we can approximate and replace $\exp\left[\frac{\chi^2_\alpha(1)}{n}\right]$ by $1+\frac{\chi^2_\alpha(1)}{n}$. So in place of $\exp\left[\frac{\chi^2_\alpha(1)}{n}\right]$ in applying the confidence interval procedure, we can use the following:

$$1+\frac{Z^2_{\alpha/2}}{\gamma} \qquad \left(\text{or } 1+\frac{Z^2_{\alpha/2}}{n}\right)$$
$$\text{or } 1+\frac{t^2_{\alpha/2}}{\gamma} \qquad \left(\text{or } 1+\frac{t^2_{\alpha/2}}{n}\right)$$
$$\text{or } 1+\frac{\chi^2_{\alpha/2}}{\gamma} \qquad \left(\text{or } 1+\frac{\chi^2_{\alpha/2}}{n}\right)$$

where γ is the degrees of freedom associated with the sum of squares due to residuals.

Transformation of explanatory variables: Box and Tidwell procedure

Suppose the relationship between y and one or more explanatory variables are nonlinear. Other usual assumptions, such as normally and independently distributed study variables with constant variance, are at least approximately satisfied.

We select an appropriate transformation on the explanatory variable so that the relationship between y and the transformed explanatory variable is as simple as possible.

The Box and Tidwell procedure describes a general analytical procedure for determining the form of transformation on x